

Research Article

High-Temperature Conductive Stability of ITO/Pt Bilayer-Film Electrode for Applications in High-Temperature SAW Devices

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Surface-acoustic-wave (SAW) devices have been widely investigated over many years as demand has grown for information sensing above 1000°C. At such high temperatures, metal electrode creep occurs, thereby causing the transformation of an electrode from a continuous uniform film to a discontinuous film with hollows or isolated grains. In this study, 100 nm thick ITO conductive oxide film is deposited on 100 nm thick Pt film to form an ITO/Pt bilayer-film composite electrode for high-temperature SAW devices. According to this way, the conductive stability of the Pt film electrode at high temperature can be greatly improved. Compared with the initial value, the electrical conductivity of single-layer 100 nm thick Pt film decreases to 65.4% with the area fraction of Pt at 64.1%. Meanwhile, the electrical conductivity of ITO/Pt bilayer-film decreases to only 95% with the area fraction of Pt at 63.1%. Then, the enhancement of high-temperature conductive stability can be attributed to the added ITO conduction channels. These findings demonstrate a potential route to design multilayer electrodes that can operate above 1000°C with promising applications in SAW devices needing to operate at high temperatures.

1. Introduction

Great efforts have been made to develop high-temperature surface-acoustic-wave (SAW) devices for many promising applications [1–7]. The propagation of acoustic waves is realized by the vibration of a piezoelectric substrate; thus, the performance of SAW devices is determined to a great extent by the mass loading effect of interdigital transducer electrodes (ITDs)—that is, too thick an electrode will decay the propagation of SAW and influence the performance of the device [8, 9]. When environmental temperatures are lower than 700°C, conventional monolayer metal films such as those of Au and Pt can be exploited as ITDs for high-temperature SAW devices without considering the deterioration of electrodes [10–13]. For SAW devices operating at

high temperatures (above 700°C and especially above 1000°C), thin metal electrodes exhibit poor conductive stability because creep-induced discontinuous holes in the metal films significantly increase their resistance [14, 15]. In contrast, $\text{La}_3\text{Ga}_5\text{SiO}_{14}$ (LGS), as a mature piezoelectric substrate, can maintain piezoelectric behavior up to its melting point at 1470°C [16, 17]. Therefore, finding electrodes that can work continuously and steadily is the most crucial factor to ensure the normal operation of LGS-based SAW devices at extremely high temperatures over long periods of time.

In order to enhance the conductive stability of film electrodes for SAW devices at high temperatures, much research has been done over many decades. Except for their important roles in nanocatalyst, [18, 19] the multilayer metal films